

Jaume **Ivars Grimalt**

Software Engineer · Backend, Data & ML

Currently in Valencia, Spain | Open to relocation to Madrid or Barcelona

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Profile

I build and operate production systems end to end: Java/Spring Boot and NestJS backends, data pipelines and cloud on GCP and the ML models they serve. Sole engineer behind Mycrospace, a computer vision product used in 35 countries.

Technical Skills

Languages	Java, TypeScript, Python, SQL
Backend	Spring Boot, NestJS, REST, Prisma, PostgreSQL, MongoDB, Redis/BullMQ, SSE/WebSocket, Stripe
Data & ML	PyTorch, ONNX, data pipelines, dataset versioning, multi-GPU training (DDP, AMP) on SLURM
Frontend & Mobile	Next.js, React, React Native/Expo
Cloud & DevOps	GCP (Cloud Run, Cloud SQL, GCS, Eventarc), Docker, CI/CD, Git, Scrum

Experience

Mycrospace — *Valencia, Spain*

Co-Founder & CTO — *Jan 2024 – Oct 2026*

- Computer vision product for microbiology labs, from Master's thesis to company; **700+** users in 35 countries; tested in **5 labs**; 1st Prize IdeasUPV (2024) and Bioval Young Talent prize. Sole engineer; managed an intern.
- **Backend.** Started the backend in Java/Spring Boot and migrated it to NestJS/TypeScript so the platform used one language. NestJS/Prisma/PostgreSQL REST API and MongoDB; workers with Redis/BullMQ and Eventarc; signed-URL uploads to GCS; real-time updates over SSE/WebSocket.
- **Infra.** Deployed on GCP (Cloud Run, Cloud SQL, GCS) with Docker and CI/CD; ONNX models in production; developed with Git and Scrum.
- **Data.** Built the image data pipeline with microbiologists: annotated **1,500** of **5,000** images and used the unlabelled data for self-supervised learning; weakly-supervised labelling from a single click per colony; dataset versioning and k-fold label scoring.
- **Frontend.** Next.js web app and React Native/Expo mobile app; Stripe payments.
- **ML.** Trained the detection model behind the product from scratch (**92%** recall / **93%** precision; **12,500+** GPU-hours on the EuroHPC Leonardo supercomputer, multi-GPU PyTorch on SLURM).

MIALAB: Medical Imaging Analysis Laboratory, UPV — *Valencia, Spain*

Research Engineer (full-time) — *May 2025 – Jan 2026*

- Profiled the volBrain pipeline (**700k+** volumes) from NIfTI ingest to voxel prediction for inference cost; volumetric brain MRI segmentation with nnU-Net.

Tapstar — *Valencia, Spain*

Full-Stack Engineer (freelance) — *Jan 2025 – May 2025*

- QR generation, back office and frontend for QR cards and stands; scan analytics and per-employee ratings.

Vesta-Z — *Valencia, Spain*

Data Engineer — *Jul 2023 – Feb 2025*

- Built full-stack features with Java/Spring Boot and PostgreSQL on GCP for construction-machinery data; designed a custom schema to cross-reference items across providers.
- Built an LLM-based customer-support bot.

Earlier experience: Neurocatching, Data Scientist Intern, 2023; Sciling, Backend Developer Intern, 2022.

Education

M.Sc. Artificial Intelligence, Pattern Recognition & Digital Imaging — *Valencia, Spain*

UPV · GPA 8.7/10 · Thesis Honours (became Mycrospace) — *2023 – 2024*

Technology Management — *Valencia / Munich*

CDTM (TUM/LMU) · 16 selected — *2024*

B.Sc. Computer Engineering — *Valencia, Spain*

UPV · GPA 8.3/10 · Honours in ML and Statistics — *2019 – 2023*

Publications & Research

- *Auto-N2N: Self-supervised MRI Denoising using Synthetic Pair Generation and Mixture of Experts.* Preprint, 2026.
- *Detecting Cytopathic Effect in Cell Culture with Computer Vision.* Under review.
- *What Extra Euler Steps Reveal About Residual Networks: Feature Dynamics in Weight-Tied and Untied Stages.* Manuscript in preparation.

Languages: Spanish (native), Catalan (native), English (C1)